# Divanshu Bhargava

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LeetCode | CodeChef | Codeforces

## Summary

Final year B.Tech CSE student at IIIT Jabalpur (2027 grad) with a strong foundation in Data Structures, Algorithms, OOP, and full-stack development at scale. Experienced in building high-performance, distributed systems with Generative AI (GenAI) and Agentic AI integrations — including an autonomous AI payment platform (1st place, IIT Roorkee Cognizance 2026), an AI-driven event orchestration platform (MongoDB Track Winner, HackByte 4.0 MLH 2026), and a 13-microservice multi-tenant university ERP (EduHub). Proficient in C++, Java, and Python with hands-on exposure to LLM orchestration, multi-agent systems, microservices, service-oriented architectures, and large-scale databases. AEH Intern at Accenture; LeetCode Knight and CodeChef 4-star competitive programmer with a self-driven, ownership-first mindset.

## Education

**Indian Institute of Information Technology** — Jabalpur, India
*B.Tech in Computer Science and Engineering — Final Year* | CPI: 7.8/10 — *Aug 2023 – May 2027*

## Experience

**Accenture** — India
*Advanced Application Engineer (AEH) Intern* — *May 2026 – Jul 2026*

## Technical Skills

**Languages**: C++, Java, Python, JavaScript, TypeScript, SQL
**Core CS**: Data Structures and Algorithms, Object-Oriented Programming, Problem Solving, DBMS, Operating Systems, Computer Networks, System Design
**Software Engineering**: Software Design, Software Development, Software Testing, Documentation, Code Review, Debugging, API Development
**Backend**: Node.js, Express.js, Django, FastAPI, Next.js (API routes), RESTful APIs, Microservices
**Databases**: PostgreSQL, MySQL, MongoDB Atlas, Query Optimization, Schema Design
**AI / ML**: Scikit-learn, TensorFlow, NumPy, Pandas, Statistical Modeling, Predictive Modeling
**GenAI & Agentic AI**: Generative AI (GenAI), Agentic AI Systems, Multi-Agent Orchestration, LLM Orchestration, RAG (Retrieval-Augmented Generation), Prompt Engineering, LangChain, LangSmith, Gemini API, Groq/LLaMA
**DevOps & Tools**: Docker, Git, JWT, RabbitMQ, PgBouncer, Redis, Celery, Prometheus, Grafana, Postman, SpacetimeDB, CI/CD (GitHub Actions), Automation, Performance Optimization, Security, QA Validation, Version Control, Unit Testing
**Data & Visualization**: Tableau, Plotly, Matplotlib, Excel
**Soft Skills**: Communication Skills, Logical Thinking, Collaborative Mindset, Quick Learner

## Achievements

- **Cognizance 2026, IIT Roorkee:** Won 1st place at Asia's second-largest student-run tech fest by creating Delivrd, an agentic AI payment and verification platform with cryptographic escrow.
- **HackByte 4.0 (MLH Official 2026):** Won MongoDB Track with ELIXA, leading all backend architecture and AI agent integration. Also placed Top 8 at HackByte 3.0 among 120 teams selected from 1,200+ nationwide participants.
- **Flipkart Grid 7.0:** Semi-finalist in one of India's largest engineering challenges, competing in advanced system design and competitive programming.
- **Competitive Programming:** LeetCode Knight, CodeChef 4-star, Codeforces Pupil — demonstrating strong DSA and algorithmic problem-solving skills.
- **Leadership:** Piloted a city-wide debate initiative involving 20+ competing teams, building stakeholder management and communication skills.

## Projects

**SalesPipe — Multi-Tenant B2B Sales CRM (Event-Driven Modular Monolith)** — 2026
**Personal Project**
**Tech Stack:** Java 21, Spring Boot 3, Spring Modulith, PostgreSQL 16, Redis, Apache Kafka, Debezium (CDC), FastAPI, Sentence-BERT, XGBoost/LightGBM, MLflow, Resilience4j, Docker, Kubernetes, Helm, KEDA, Argo CD, Prometheus/Grafana, OpenTelemetry, Next.js 15

- Designing a multi-tenant B2B Sales CRM as a modular monolith spanning 10 domain modules (identity, CRM core, pipeline, activity, email tracking, scoring, notification, eventing, reporting) with compile-time boundary enforcement via Spring Modulith + ArchUnit, deployed on Kubernetes with HPA autoscaling across 3–10 pods. GitHub
- Architected a transactional-outbox event backbone (Debezium CDC → Kafka) spanning 5 domain event topics with Schema Registry-enforced compatibility and idempotent, dedupe-store-backed consumers; a chaos test recovering a 300-event burst mid-crash processed all 300 events exactly once with zero loss or duplication.
- Building an AI lead-scoring pipeline fusing Sentence-BERT (384-dim) embeddings with structured engagement/firmographic features into an XGBoost/LightGBM classifier served via FastAPI behind a Resilience4j circuit breaker, with MLflow-managed versioning and auto-promotion gated on AUC-ROC and precision@k.
- Modeled an 18-table multi-tenant schema with Hibernate row-level tenant isolation, rotating refresh tokens with reuse detection, and HMAC-signed email-tracking tokens; planning Prometheus/Grafana/OpenTelemetry observability and a 2-chart Helm + Argo CD (GitOps) + KEDA deployment pipeline.

**EduHub — Multi-Tenant SaaS University ERP Platform** 2026

**Capstone Project, Ongoing**

**Tech Stack:** Django 5 + DRF, FastAPI, PostgreSQL 16, PgBouncer, Redis, Celery, RabbitMQ, JWT (HS256), VADER, Docker, Prometheus/Grafana, Next.js, Tailwind CSS, GitHub Actions (CI/CD)

- Architected a multi-tenant SaaS ERP as 13 independently deployable microservices (Django REST Framework + FastAPI) across 7 fully-implemented domains (auth, finance, hostel, transport, grievance, notification, AI) plus 6 stubbed domains, onboarding unlimited universities on one common database/schema. GitHub
- Designed an event-driven backbone on a RabbitMQ topic exchange routing 9+ domain event types with a transactional outbox, idempotent inbox, and dead-letter queuing, achieving exactly-once-effective delivery with zero cross-service synchronous coupling.
- Secured the platform with zero-trust JWT (HS256) independently re-verified by all 13 services and row-level tenant isolation; delivered 2 production-grade distributed sagas, including a 3-service hostel-payment compensation saga and an ML-powered grievance auto-escalation pipeline (VADER sentiment scoring).
- Validated reliability with 143+ backend tests (≥70% coverage gate) and 19 frontend tests via a 4-stage CI/CD pipeline; standardized API responses (`{success, data, message, errors}`, paginated 20/100) with Prometheus/Grafana observability across all services.

**Delivrd — Agentic AI Payment & Project Orchestration Platform** 2026

**1st Place, IIT Roorkee Cognizance 2026**

**Tech Stack:** FastAPI (Python 3.12), PostgreSQL (async SQLAlchemy 2, asyncpg), Groq LLaMA 3.3-70B, React 19, Vite, Tailwind CSS, JWT + bcrypt, SHA-256 / HMAC-SHA256, Docker, SonarQube, Figma API

- Developed an agentic AI freelance platform that autonomously decomposes projects into 3–9 milestones via LLaMA 3.3-70B (Groq), with modality-aware verification (code, content, design) and cryptographically secured escrow payments. GitHub | Demo
- Assembled a 4-layer code verification pipeline (AST -> Docker sandbox -> SonarQube -> LLM review) and a 7-step content + 5-dimension design pipeline; deployed a SHA-256 chain-hashed escrow ledger with HMAC-SHA256 signed, idempotency-keyed financial operations.

**ELIXA — GenAI-Powered Agentic Event Orchestration Platform** 2026

**MongoDB Track Winner, HackByte 4.0(MLH)**

**Tech Stack:** Next.js 14, SpacetimeDB (Rust), MongoDB Atlas, Gemini 2.5 Flash, Claude Sonnet 4, ElevenLabs Turbo v2, iron-session, Tailwind CSS + shadcn/ui, Framer Motion

- Created an agentic event coordination platform converting plain-language descriptions into 30–60 dependency-linked tasks across 6 phases using Gemini 2.5 Flash, with phase-gated checkpoints and ElevenLabs Turbo v2 voice announcements. GitHub | Demo | Live
- Architected the full backend: 5 SpacetimeDB live-state tables, all Next.js 14 API routes, and a 3-layer persistence system (SpacetimeDB -> HTTP relay -> MongoDB fallback) for resilient session handling.
- Optimized real-time task sync to sub-300ms latency across 6 role-scoped operator types; established a dependency-locked Go/No-Go launch gate requiring 100% critical-task completion across all 6 phases before event kickoff.